INCH-POUND

MIL-M-38510/110B
AMENDMENT 1
01 DECEMBER 1994

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, QUAD OPERATIONAL AMPLIFIERS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/110B, dated 11 September 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, add footnote for device type 01 which will state "Devices may be monolithic or they may consist of two separate independent die."

PAGE 2

3/, second line, delete the words inside the parenthesis, and substitute "(36 volts for device types 05 and 06)."

PAGE 3

3.1, delete and substitute the following;
"3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 4

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Input offset voltage test. Footnote "1/" will be renumbered as footnote "2/".

TABLE I, Input offset current test. Footnote "1/" will be renumbered as footnote "2/".

TABLE I. Input bias current test. Footnote "1/" will be renumbered as footnote "2/". Device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min". Device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

TABLE I, Input voltage common mode rejection test. Footnote "2/" will be renumbered as footnote "3/".

TABLE I. Output short circuit current (for positive output) test. Footnote "3/" will be renumbered as footnote "4/". Footnote "4/" will be renumbered as footnote "5/".

TABLE I. Output short circuit current (for negative output) test. Footnote "3/" will be renumbered as footnote "4/". Footnote "4/" will be renumbered as footnote "5/".

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PAGE 5

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Supply current test, footnote "5/" will be renumbered as footnote "6/".

TABLE I, Single ended open loop voltage gain test, footnote "6/" will be renumbered as footnote "7/".

PAGE 6

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."
TABLE I, Transient response test, footnote "7/" will be renumbered as footnote "8/".

PAGE 7

Add a new footnote "1/" which will state;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Footnote "1/" will be renumbered as footnote "2/". At the second sentence, delete the words "Device type 05" and substitute "Device types 05 and 06".

Footnote "2/" will be renumbered as footnote "3/".

Footnote "3/" will be renumbered as footnote "4/".

Footnote "4/" will be renumbered as footnote "5/".

Footnote "5/" will be renumbered as footnote "6/".

Footnote "6/" will be renumbered as footnote "7/".

Footnote "7/" will be renumbered as footnote "8/". On the first sentence, delete the words "Device type 05" and substitute "Device types 05 and 06".

PAGE 14

FIGURE 3, add the word "minimum" after "T_A = +125°C". Delete "$+V_{IN}$" and substitute "$\pm V_{IN}$".

PAGE 15

FIGURE 4, add the word "minimum" after "T_A = +125°C".

PAGE 24

Test numbers 9-16, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 9-16, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

Test number 23, adapter pin number 2, delete "5 V" and substitute "-5 V".

PAGE 25

Test numbers 33-40, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 33-40, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

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PAGE 26

Test numbers 57-64, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 57-64, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

PAGE 35

1/, add the following sentence; "For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

PAGE 37

4.1, delete and substitute the following;
"4.1 Sampling and inspection. Sampling inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 38

TABLE IV, $+I_{IB}$ delta limits for device type 04, delete "-25 nA max" and substitute "+25 nA max".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
DLA - ES

Agent:
DLA - ES

(Project 5962-1403)

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